

Sentiment Analysis of Amazon Customer Reviews Using NLP

1. Introduction

Sentiment Analysis is a Natural Language Processing (NLP) technique used to determine the emotional tone of textual data. It helps organizations understand customer opinions, feedback, and attitudes toward products or services. In this project, sentiment analysis was performed on Amazon customer reviews to classify them as Positive, Negative, or Neutral.

2. Objective

To analyze customer review text data, classify reviews as Positive, Negative, or Neutral, apply NLP techniques, understand public opinion through sentiment patterns, and provide insights useful for marketing and product development.

3. Dataset

The dataset used for this project was the Amazon Reviews Dataset obtained from Kaggle. A sample of 5,000 reviews was selected for analysis.

	label	title	text
0	2	Stuning even for the non-gamer	This sound track was beautiful! It paints the ...
1	2	The best soundtrack ever to anything.	I'm reading a lot of reviews saying that this ...
2	2	Amazing!	This soundtrack is my favorite music of all ti...
3	2	Excellent Soundtrack	I truly like this soundtrack and I enjoy video...
4	2	Remember, Pull Your Jaw Off The Floor After He...	If you've played the game, you know how divine...

4. Tools and Technologies Used

Python, Kaggle Notebook, Pandas, TextBlob, and Matplotlib.

5. Methodology

The dataset was loaded into a Pandas DataFrame. TextBlob was used to calculate sentiment polarity scores. Reviews were classified as Positive, Negative, or Neutral. Sentiment frequencies were counted and visualized using a pie chart.

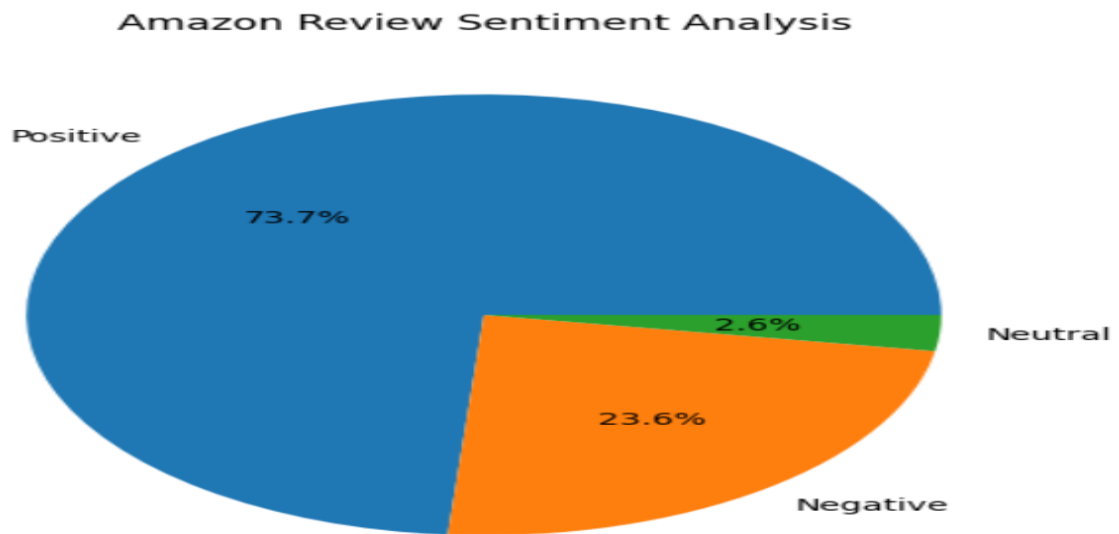
6. Sentiment Analysis Output

	label	title	text	Predicted_Sentiment
0	2	Stuning even for the non-gamer	This sound track was beautiful! It paints the ...	Negative
1	2	The best soundtrack ever to anything.	I'm reading a lot of reviews saying that this ...	Positive
2	2	Amazing!	This soundtrack is my favorite music of all ti...	Positive
3	2	Excellent Soundtrack	I truly like this soundtrack and I enjoy video...	Positive
4	2	Remember, Pull Your Jaw Off The Floor After He...	If you've played the game, you know how divine...	Positive

7. Sentiment Count Results

```
Predicted_Sentiment
Positive      3687
Negative     1182
Neutral       131
Name: count, dtype: int64
```

8. Pie Chart Visualization



9. Results and Analysis

Positive Reviews: 73.7%, Negative Reviews: 23.6%, Neutral Reviews: 2.6%. The majority of reviews were positive, indicating high customer satisfaction and favorable customer opinions.

10. NLP Technique Used

This project used Lexicon-Based Sentiment Analysis through the TextBlob library. TextBlob calculates polarity scores and classifies reviews according to sentiment.

11. Conclusion

This project successfully applied NLP techniques to analyze Amazon customer reviews. Most reviews were positive, indicating strong customer satisfaction. The findings can help businesses improve products, marketing strategies, and customer experience.